Quantification: 699_HYP_HP_z20c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1438.44	1239	799.765	px^2
major_axis_length_px	lacuna shape	70.1037	63.7	22.8233	px
minor_axis_length_px	lacuna shape	33.4837	30.7419	15.9453	px
aspect_ratio	lacuna shape	2.5837	1.665	1.5667	unitless
eccentricity	lacuna shape	0.8208	0.7995	0.1366	unitless
solidity	lacuna shape	0.7565	0.7599	0.117	unitless
roots_count	headline	10.6667	8	13.2688	count
ring_length_r30_px	headline	395.083	305.5	398.719	px
ring_length_r60_px	reported	734.833	670.5	633.388	px
owned_length_px	ownership-dependent	899.257	803.633	831.698	px
edge_count	ownership-dependent	50.3333	45	50.5071	count
mean_edge_length_px	ownership-dependent	19.9754	18.6778	4.2761	px
perimeter_px	normalised	323.502	189.965	362.545	px
ring_area_r30_px2	normalised	9610.67	7018	6578.44	px^2
ring_area_r60_px2	normalised	21960.1	17966.5	11085.4	px^2
in_frame_fraction_r30	normalised	0.9802	1	0.0685	unitless
in_frame_fraction_r60	normalised	0.9722	1	0.0964	unitless
ring_density_r30	normalised	0.0382864	0.0374402	0.0180917	px^-1
ring_density_r60	normalised	0.031436	0.0360087	0.0157673	px^-1
roots_per_100px_perimeter	normalised	3.2818	3.2885	1.6559	per 100 px
ring_attached_length_r30_px	network v2	351.917	226	393.472	px
ring_attached_length_r60_px	network v2	574.5	355	621.692	px
ring_length_w_r30_px	network v2	441.517	338.794	450.03	px
ring_length_w_r60_px	network v2	825.295	751.293	721.046	px
sholl_crossings_r10	network v2	8.4167	5.5	9.709	count
sholl_crossings_r20	network v2	9.4167	7.5	7.8214	count
sholl_crossings_r30	network v2	10.8333	10.5	8.6006	count
loop_count	loops, report only	1.9167	1	2.7784	count
loop_edge_share	loops, report only	0.087	0.0557	0.0957	share
ring_width_mean_r30_px	width	6.9707	6.9965	0.9338	px

Per field
----

measure	value	units
analysed_area_px2	877006	px^2
total_skeleton_length_px	16520	px
canalicular_length_density_per_px	0.0188368	px^-1
junction_count	321	count
junction_density_per_px2	0.00036602	px^-2
lacuna_count	12	count

lacunae_per_px2	1.368e-05	px^-2
skeleton_component_count	299	count
mean_component_length_px	55.2508	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0211004	px^-1
field_density_without_flagged_per_px	0.0146526	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	952	px
unattached_component_count	47	count
canalicular_length_density_attached_per_px	0.0188368	px^-1
field_length_density_with_unattached_per_px	0.0199223	px^-1
excluded_area_px2	147318	px^2
analysed_field_area_px2	901258	px^2
loop_count	27	count
loop_edge_share	0.0978	share
edge_count	1043	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.0458	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	14362	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
2	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
3	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
4	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
5	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
6	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
7	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
8	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
9	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False
10								283.342	208.259	False
11								341.9	874.582	False
12								546.189	775.469	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	251	758	1072.15	56	19.1456	8.1471
2	8	355	768	954.578	57	16.747	6.9021
3	9	418	827	1308.66	72	18.1758	7.1351
4	8	273	559	580.961	30	19.3653	6.0822
5	4	338	583	652.688	39	16.7356	7.8428

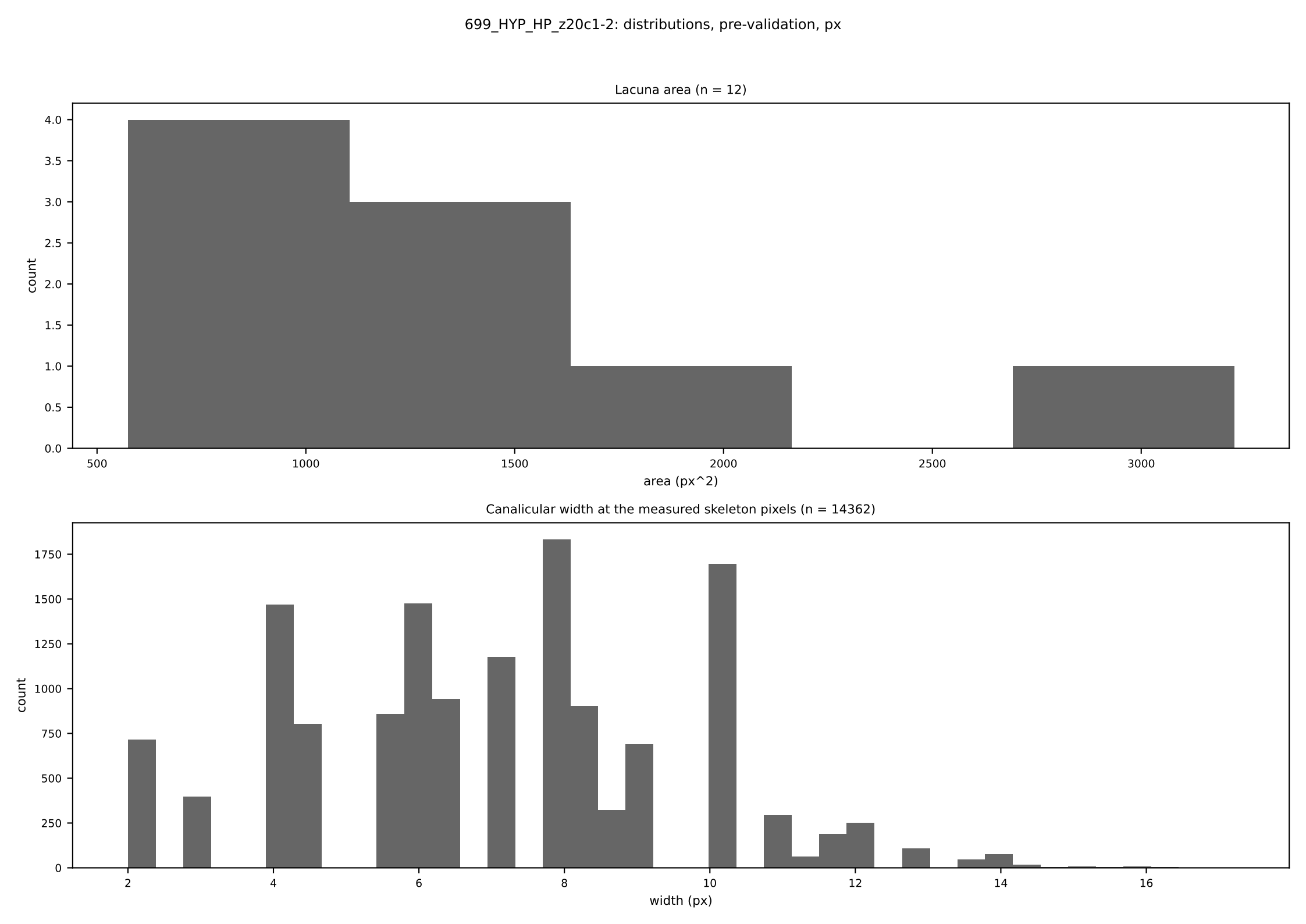
6	15	566	990	984.607	51	19.306	7.8473
7	8	416	892	1329.33	73	18.21	6.6009
8	11	199	375	258.924	17	15.2308	5.9147
9	2	36	63	53.1838	2	26.5919	5.3491
10	51	1576	2521	3140.54	190	16.5291	8.2752
11	2	123	234	265.966	10	26.5966	6.4606
12	3	190	248	189.498	7	27.0711	7.0909

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	185.581	7275	19284	1	1	0.0345017	0.0393072	3.7719
2	418.517	11182	24664	1	1	0.0317475	0.0311385	1.9115
3	221.179	7585	17507	1	1	0.0551088	0.0472383	4.0691
4	140.326	6761	16630	1	1	0.0403786	0.0336139	5.701
5	165.74	6360	13932	1	1	0.0531447	0.0418461	2.4134
6	316.149	10460	25779	1	1	0.0541109	0.0384034	4.7446
7	132.267	6479	17177	1	1	0.0642074	0.0519299	6.0484
8	363.747	11289	26807	1	1	0.0176278	0.0139889	3.0241
9	163.681	5591	13507	0.7628	0.6662	0.00643892	0.00466425	1.2219
10	1435.5	29498	54199	1	1	0.0534273	0.0465138	3.5528
11	144.995	6436	18426	1	1	0.0191112	0.0126994	1.3794
12	194.35	6412	15609	1	1	0.0296319	0.0158883	1.5436

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	213	663	280.38	845.784	4	7	11
2	216	332	401.505	860.423	9	9	12
3	322	508	464.049	922.269	6	11	10
4	236	355	305.723	629.73	5	8	8
5	258	355	371.865	656.801	7	7	11
6	564	785	646.428	1131.97	15	13	12
7	411	812	461.321	999.24	6	13	15
8	169	244	216.439	405.321	5	3	6
9	24	24	41.0416	71.4264	0	0	1
10	1520	2393	1772.46	2858.8	37	31	35
11	109	194	133.012	256.581	3	4	5
12	181	229	203.983	265.196	4	7	4


699_HYP_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 14362)
count
width (px)